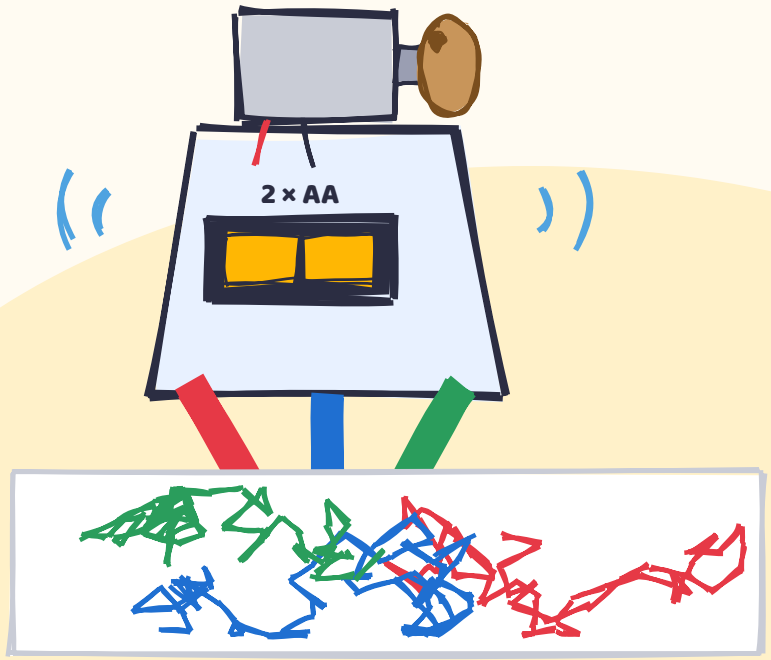
Junior Makers · Introduction

Welcome to Robotics!

What robots are, how they work, and your first build: an Art Bot

Sense, think, act | Parts of a robot | DC motor + vibration

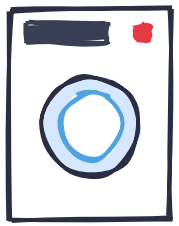


Did you know?
Your phone vibrates using a tiny motor with an off-centre weight. Today you use the same trick to make art!

Quick quiz

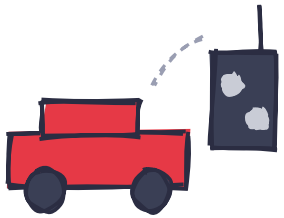
Is it a robot? Thumbs up or down!

1



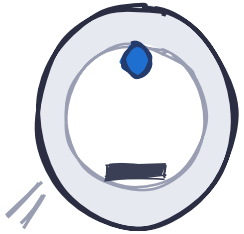
Washing machine

2



Remote-control car

3



Robot vacuum

4



Calculator

5



Self-driving car

6



Delivery drone

? Talk with your partner: **what makes something a robot** and not just a machine?

The big idea

A robot can SENSE, THINK and ACT



SENSE

It notices the world with **sensors**



THINK

Its **brain** decides what to do



ACT

It moves or reacts with **actuators**



REPEAT

Again and again, by itself

Our definition

A robot is a **machine** that can **sense** its surroundings, **decide** what to do using a **program**, and then **act**, all **by itself**.

You do it too!

You SEE a ball coming, your brain THINKS 'catch!', your hands ACT.

Watch and think

Video time!



How to Build an ArtBot

Science Buddies · about 2 minutes · youtu.be/daWU2Oh_xlg

Watch for...

1. What makes the ArtBot **move**?
2. Why does it wobble instead of driving straight?
3. How could you change the **pattern** it draws?

Scan to watch

After the video: Predict: if you move the weight on the motor, will the drawing change? How?

Answers

So... which ones are robots?

Sort of — Washing machine
It follows a program and senses water level, but it cannot move around or change plans much.

No — Remote-control car
A person does all the thinking. No sensing, no deciding by itself.

Yes! — Robot vacuum
Senses walls and dirt, decides where to go, moves by itself.

No — Calculator
It thinks (calculates) but cannot sense the world or move.

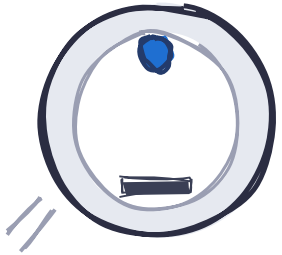
Yes! — Self-driving car
Cameras and radar sense, a computer decides, motors steer and brake.

Yes! — Delivery drone
Uses GPS and sensors to fly a route and land on its own.

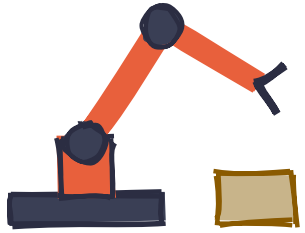
The more a machine can **sense, decide and act on its own**, the more "robot" it is. Engineers call this being **autonomous**.

Look around you

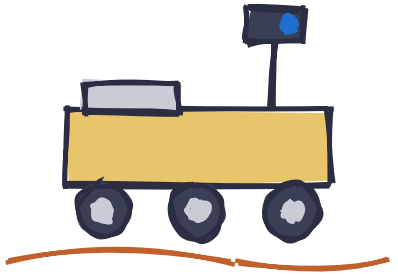
Robots are everywhere!



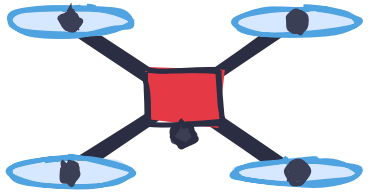
Home helper
Cleans floors by itself



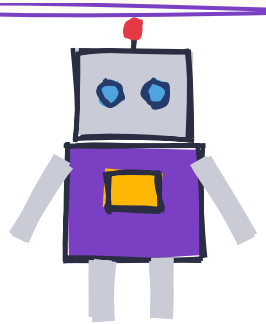
Factory arm
Builds cars, packs boxes



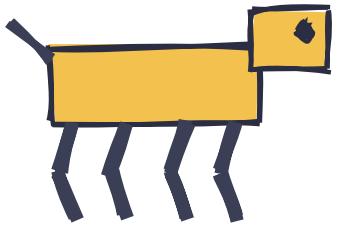
Mars rover
Explores another planet



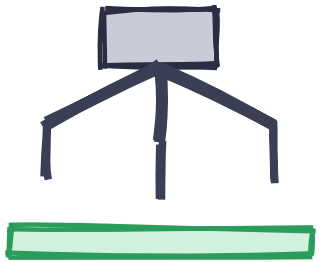
Drone
Films, maps, delivers



Humanoid
Walks and talks like us



Robot dog
Inspects dangerous places



Surgical robot
Helps doctors operate



Self-driving car
Drives itself on roads

A short history

From an idea to everywhere

1921

The word **robot** first appears in a Czech play by Karel Čapek. It comes from *robota*, meaning hard work.

1961

Unimate, the first factory robot arm, starts work in a car factory in the USA.

1997

Sojourner, a rover the size of a microwave, drives on Mars.

2002

Roomba robot vacuums go on sale and robots enter our homes.

2021

Ingenuity makes the first powered flight on another planet, Mars.

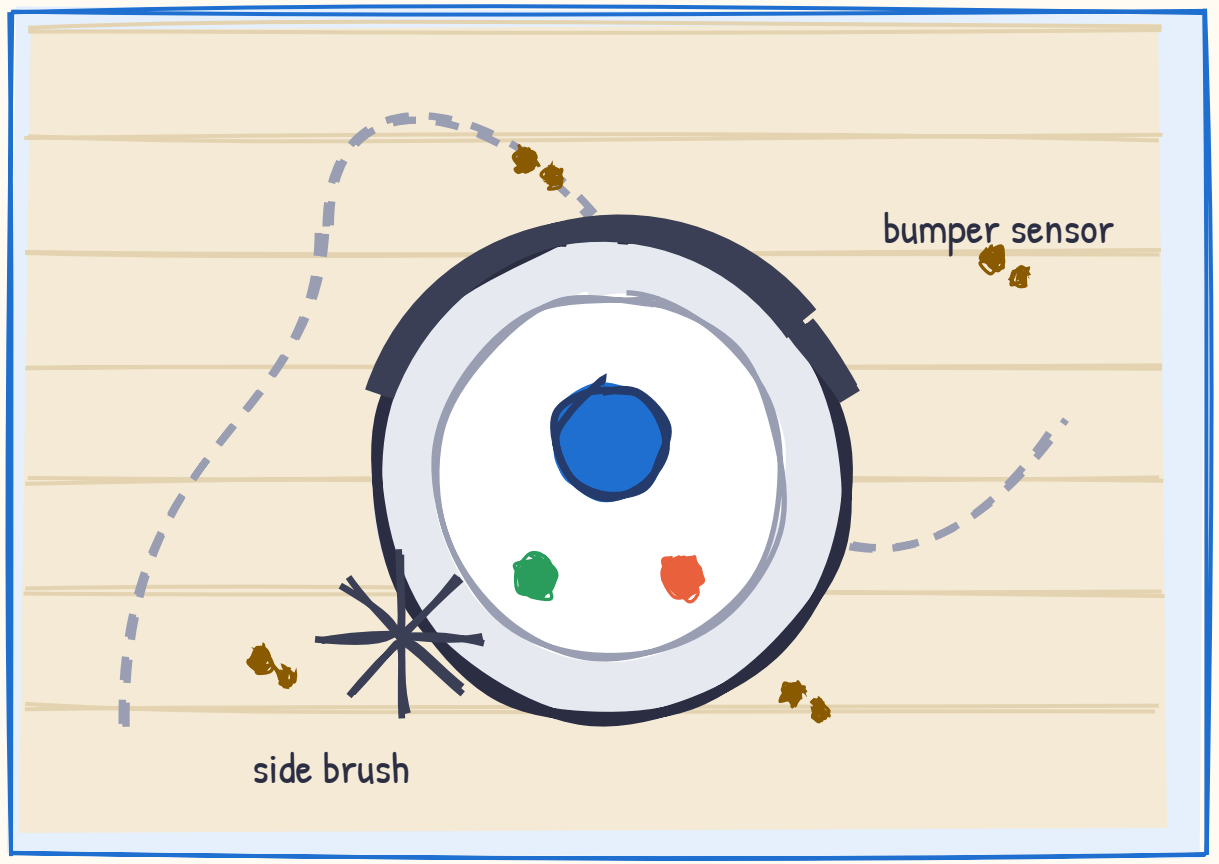
Today

Robots work in hospitals, farms, warehouses and space, often with **AI** inside.

LEARN

Robot of the home · iRobot, USA · 2002

Roomba



What it does

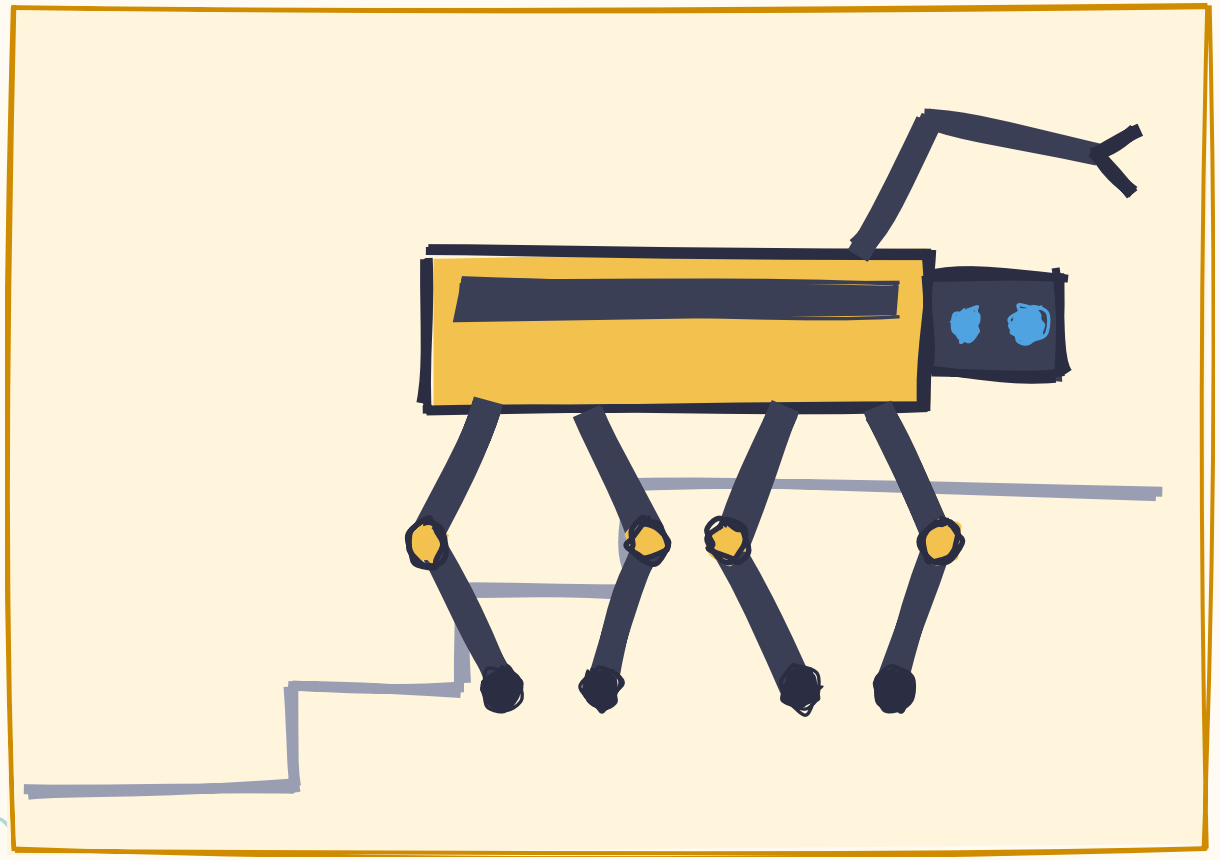
A round robot that **vacuums your floor by itself**. It drives around furniture, cleans under beds and goes back to its **charging dock** when its battery is low.

SENSE: bump sensors, cliff sensors (stop falls down stairs), dirt sensors
THINK: plans a cleaning path
ACT: wheels, spinning brushes and a vacuum

Fascinating fact
iRobot has sold tens of millions of robot vacuums. Many homes have a robot before they have any other robot!

Robot dog · Boston Dynamics, USA

Spot



What it does

A four-legged robot that **walks, climbs stairs and opens doors**. It inspects places that are dangerous for people, like power stations, oil rigs and building sites.

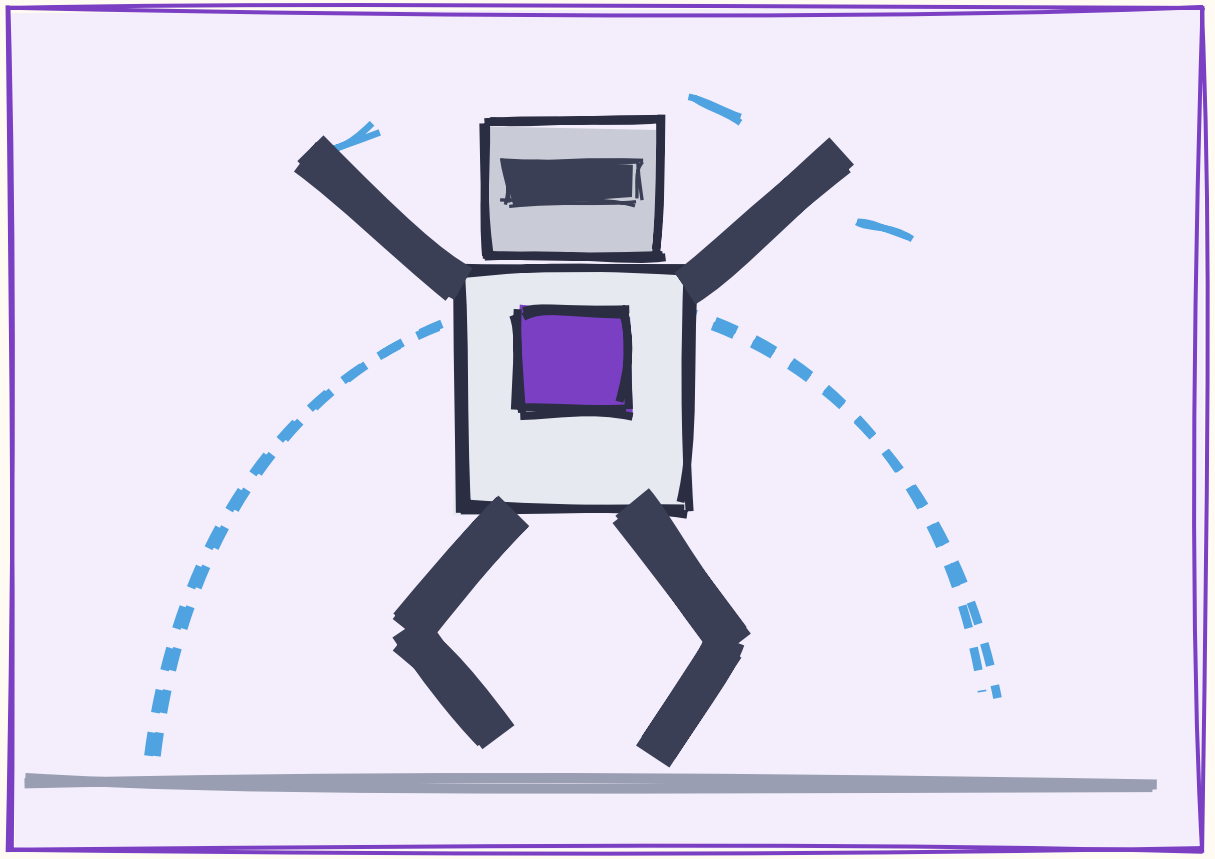
SENSE: cameras on every side see in 3D all around it
THINK: keeps its balance hundreds of times every second
ACT: 12 motors move its 4 legs; an optional arm can grab things

Fascinating fact

Spot can carry about 14 kg on its back and walk on rough ground where wheels would get stuck.

Humanoid robot · Boston Dynamics, USA

Atlas



What it does

A robot shaped like a **human** that can run, jump and even do **backflips**. Engineers use it to learn how robots can work in places built for people.

SENSE: cameras and depth sensors map the ground in front of it

THINK: a computer plans every jump and landing

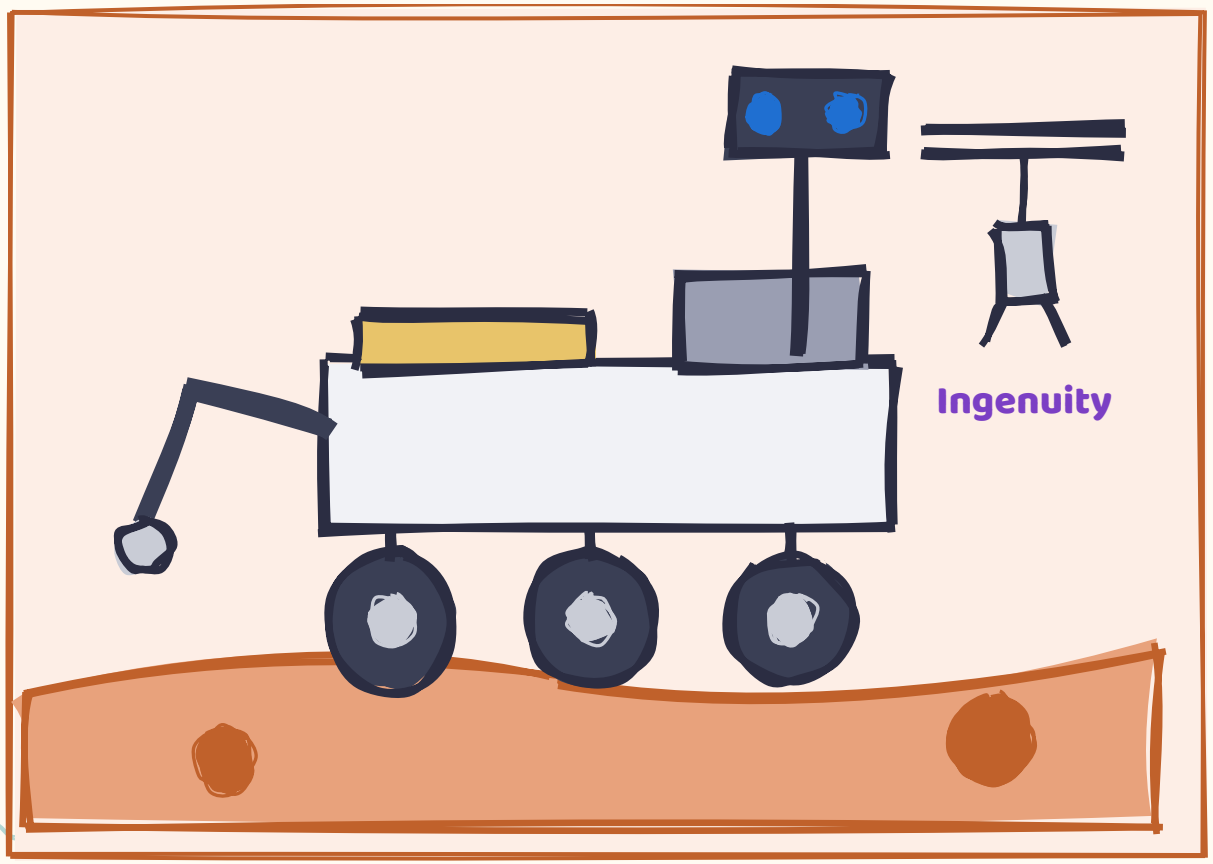
ACT: powerful motors in its joints move like our muscles

Fascinating fact

Atlas videos doing parkour have been watched by millions of people. Balancing on two legs is one of the hardest problems in robotics!

Mars rover · NASA, USA · landed 2021

Perseverance



What it does

A car-sized robot **exploring Mars**. It drills rocks, collects samples and searches for signs of ancient life. It also carried **Ingenuity**, a small helicopter that flew 72 times on Mars.

SENSE: about two dozen cameras, a laser and science instruments
THINK: drives itself between targets, because radio messages take minutes to reach Earth
ACT: 6 wheels, a robotic arm with a drill, and sample tubes

Fascinating fact
Perseverance weighs about 1,000 kg and is powered by a nuclear battery, not solar panels.

LEARN

Inside every robot

The 6 parts of a robot

Sensors = the eyes, ears and skin
Collect information

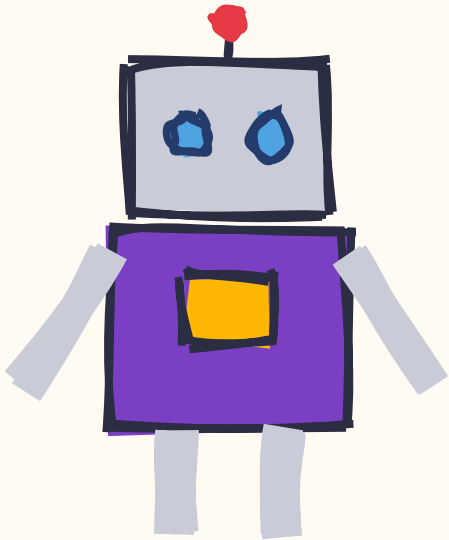
Controller = the brain
Runs the program, makes decisions

Actuators = the muscles
Move, light up, make sound

Power = the food and energy
Batteries or a power supply

Body = the skeleton
Holds everything together

Program = the thoughts
Instructions written by us



Human vs robot
We have the same 6 parts! Our nerves are the wires.

Sensing

Sensors: how robots notice the world

Ultrasonic
Distance, using sound echoes

Light (LDR)
How bright or dark it is

Camera
Pictures, faces, colours

Touch / button
Being pressed or bumped

Temperature
How hot or cold it is

Motion (PIR)
A person moving nearby



Our Art Bot today has **no sensors** at all. Remember this: at the end we will ask if that means it is **not really a robot**.

Your sensors
Eyes, ears, nose, tongue, skin. Which one is like a camera?

Acting

Actuators: how robots move and react

DC motor
Spins wheels and fans

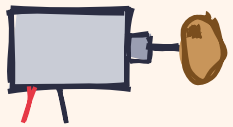
Servo motor
Turns to an exact angle

LED
Lights up to show a signal

Buzzer
Beeps an alarm or a tune

Screen
Shows words and numbers

Pump
Moves water



Today our Art Bot uses one actuator: a **DC motor**. It turns electricity into **spinning**, and we will turn that spinning into **shaking**!

Think
Which actuators does a robot vacuum need?

Thinking

The brain: a microcontroller



What is it?

A **microcontroller** is a tiny computer on a board. The **Arduino Uno** reads sensors, runs our program and controls actuators. It is cheap, safe (5 volts) and used by millions of learners.

How do we program it?

1. Write the program on a laptop.
2. **Upload** it through a USB cable.
3. The Arduino runs it **by itself**, even without the laptop.

Bigger brain

Later we use the ESP32: a brain with Wi-Fi and Bluetooth built in!

Robots + AI

When robots learn

A normal robot

Follows **rules we write**.

IF something is closer than 20 cm, THEN beep.

A robot with AI

Learns from examples.

Show it 1,000 pictures of cats and it learns to *recognise* a cat by itself.



Face unlock on phones



Cars that read road signs



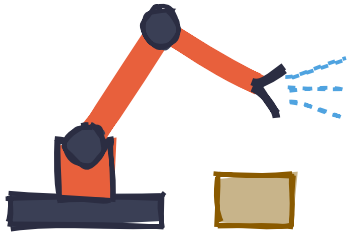
Farm robots that pick ripe fruit



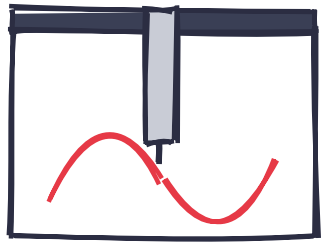
Voice assistants that understand you

Robots + art

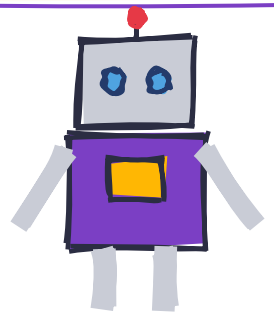
Machines that make art



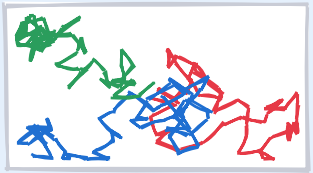
Painting arms
Spray-paint every new car in the factory, smooth and even



Drawing plotters
Pens on motors draw maps, posters and designs



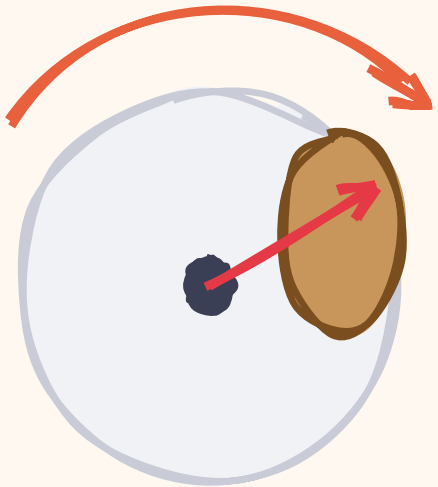
Robot artists
Some robots use cameras and AI to draw portraits



Today you build the **simplest art machine of all**: it has no brain and no plan, but its **wobble** makes a different drawing every time.

The secret

Why does it shake?



heavy side pulls it round

Off-centre weight

If the cork sits **in the middle** of the shaft, it spins smoothly: no shake. If it sits **to one side**, the heavy side pulls the motor one way, then the other way, many times every second. That pull is a **vibration**.

Your **phone** has the same thing inside! A tiny motor with an off-centre weight makes it **vibrate** when someone calls.

Try it

Hold a running motor gently. Feel the buzz in your fingers!

Big question

So… is our Art Bot a robot?

Think about it

SENSE? No sensors. It cannot notice the edge of the paper.

THINK? No brain. It cannot decide anything.

ACT? Yes! A motor moves it.

Our answer

Not yet! It is a clever **machine**: it only ACTS. To become a robot it needs to **sense** and **think** too.

How could we make it a robot?

Add a **light sensor** so it stops in the dark · add a **tilt or bump sensor** so it turns off when it reaches the edge of the paper · add a **tiny brain** (micro:bit or Arduino) to make it shake in patterns.



Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Program a human robot
2. Make an Art Bot!
3. Our builder's kit (per pair)
4. Make the shaker (motor + cork)
5. Make the body (cup + marker legs)
6. Put it together and switch on
7. Experiment: change ONE thing

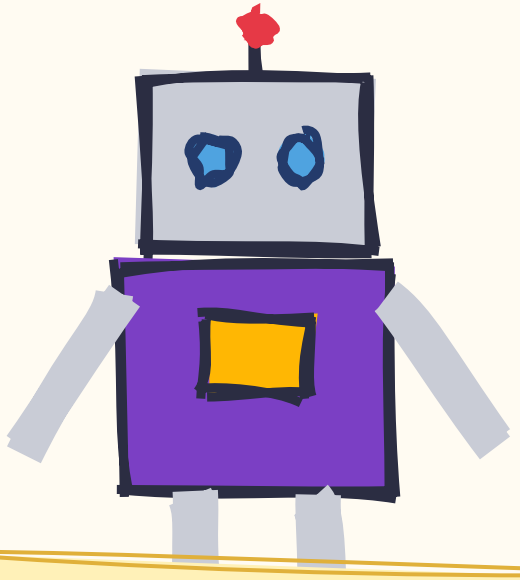
MAKE

Stand up and play!

Program a human robot

How to play (pairs)

1. One partner is the **robot**, one is the **programmer**.
2. The robot must reach a target (a chair, a door) and pick up an object.
3. The robot follows instructions **exactly**, nothing more: "step forward", "turn right", "stop".
4. If the program is wrong, the robot does the wrong thing!



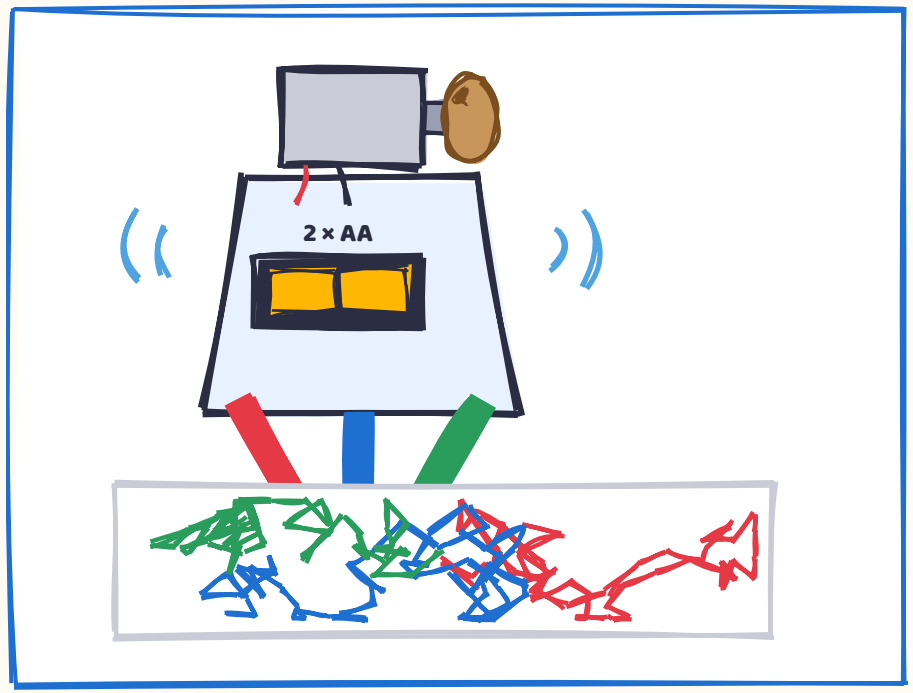
Robot rule
Robots do what we TELL them, not what we MEAN!

? Was your program right the first time? Fixing mistakes in a program is called **debugging**.

MAKE

Your build

Make an Art Bot!



How it works

A **cup** stands upside down on **marker legs**. A **DC motor** on top spins a **cork stuck on off-centre**. The heavy side swings round and round and makes the whole bot **shake**. The shaking makes it walk, and the markers **draw its path**.

Sense, think, act?

Sense: … nothing yet!

Think: … no brain yet!

Act: the motor shakes it and the markers draw

Keep this in mind

Is our Art Bot really a robot? We will decide at the end.

Build it

Our builder's kit (per pair)



Plastic or paper cup
The body



3 or 4 markers
The legs that draw



Small DC motor
Makes the shake



Cork (or eraser)
The off-centre weight



2 × AA + holder
The power



Tape + switch
Hold it all together

Safety rules

1. Use a **plastic or paper cup, never a real glass**: a shaking bot can fall off the table.
2. **Only the teacher uses the hot glue gun.**
3. Keep fingers and hair away from the spinning cork.
4. Switch off or unclip the battery whenever you change something.

Washable!
Use washable markers and cover the table with big paper.

MAKE

Build step 1 of 3

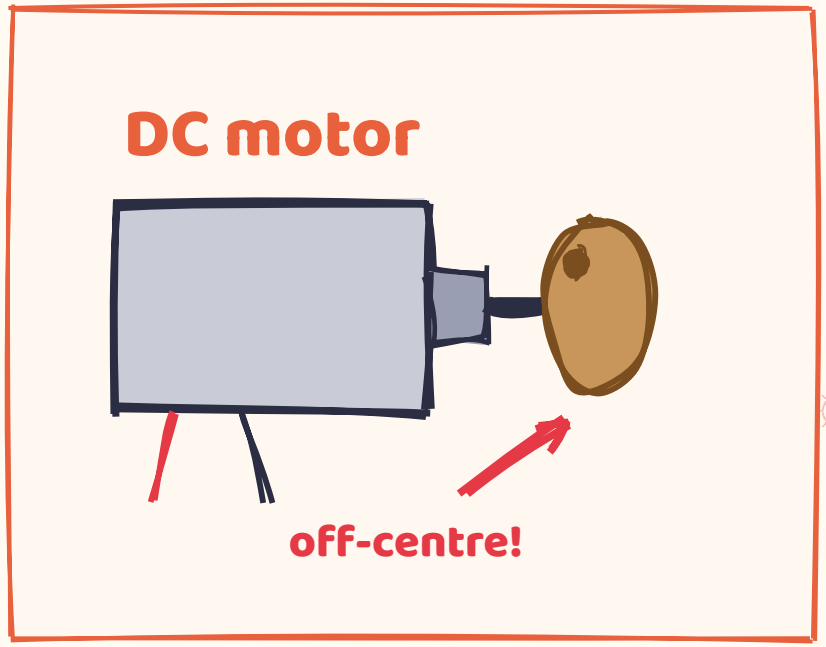
Make the shaker (motor + cork)

a Push the motor shaft into the cork **near its edge**, not in the middle. (Teacher can make a small hole first.)

b Check the cork is **tight**: it must not slip on the shaft.

c Connect the 2 motor wires to the battery holder. The cork should spin and the motor should **buzz** in your hand.

d Disconnect again before the next step.



Off-centre = shake
If the cork is in the middle, your bot will barely move. Move it to the edge!

MAKE

Build step 2 of 3

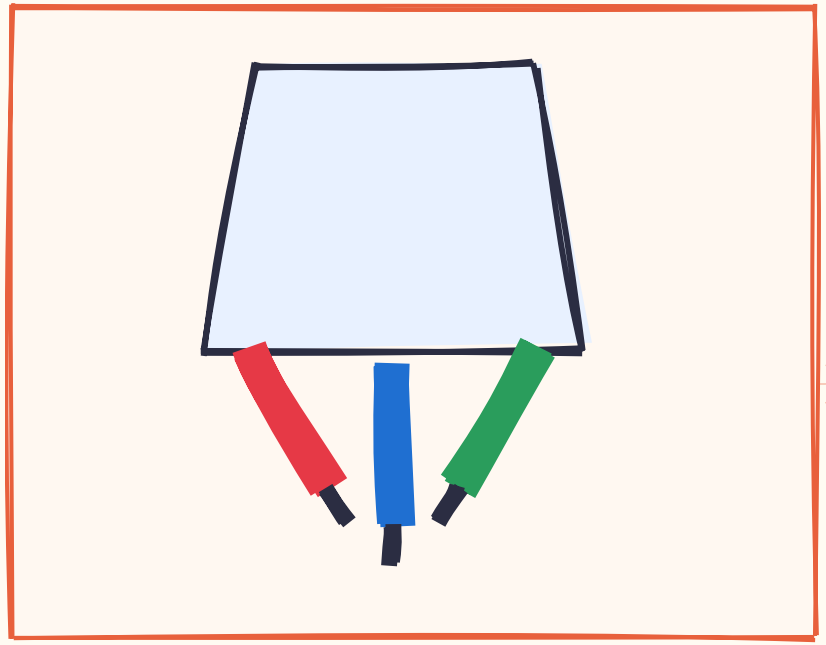
Make the body (cup + marker legs)

a Turn the cup **upside down**.

b Tape 3 or 4 markers around the rim like **legs**, tips pointing down. Spread them out evenly.

c Take the **caps off**. Make sure all tips touch the paper when the cup stands up.

d The cup should stand by itself. If it falls, adjust the leg angles.



Engineer's eye

Legs spread wide = steady bot. Legs close together = wild, wobbly bot. Which do you want?

Build step 3 of 3

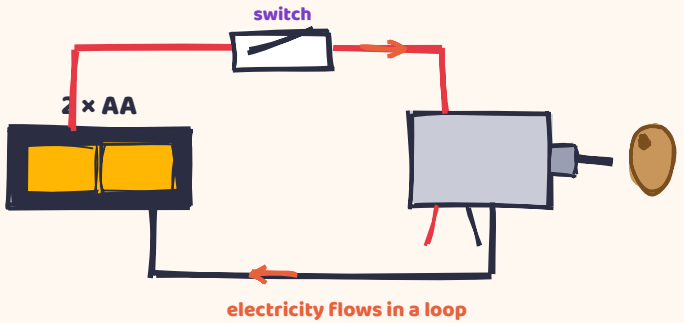
Put it together and switch on

a Teacher hot-glues (or you tape) the **motor on top** of the cup, with the cork hanging over the edge so it can spin freely.

b Tape the **battery holder** on the cup's side, balanced on the other side from the motor.

c Add a **switch** in one wire, or just clip the battery on and off.

d Place it on the big paper and switch on. **Watch it draw!**



Not moving?
Is the circuit a complete loop? Is the cork touching the cup? Are the batteries the right way round?

Test like an engineer

Experiment: change ONE thing

What we changed	How it moved	What it drew
Move the cork nearer the middle	fast / slow / spins	circles / lines / zig-zag
Move the cork to the very edge	fast / slow / spins	circles / lines / zig-zag
Legs spread wide vs close	fast / slow / spins	circles / lines / zig-zag
Add a coin on one side of the cup	fast / slow / spins	circles / lines / zig-zag
Use 3 legs vs 4 legs	fast / slow / spins	circles / lines / zig-zag

Fair test

Change **only one thing** each time. Keep the same batteries, paper and markers. Then you know what caused the change.

Name your style

Is your bot a Scribbler, a Spinner or a Zig-Zagger?

? Can you make your Art Bot draw **big circles**? Can you make it go in a **straight-ish line**? What did you change?

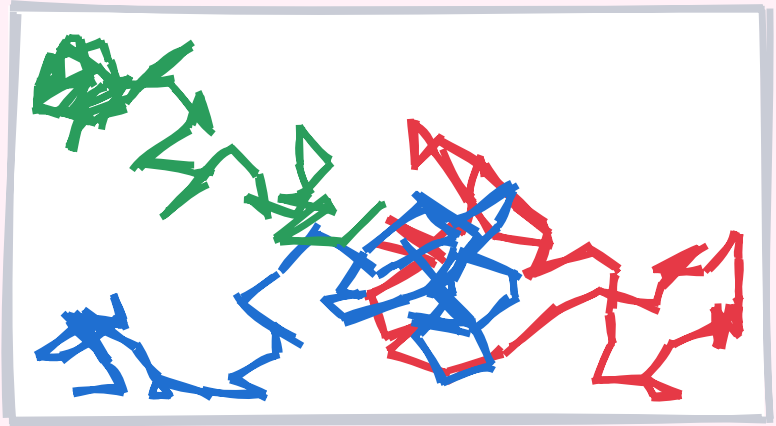
Level up!

Challenges and art gallery

Level 1
Make your bot draw for 30 seconds without falling over.

Level 2
Design a bot that draws mostly circles, then one that draws lines.

Level 3
Decorate it as a character: give it eyes, a name and a story!



Gallery walk: put every drawing on the wall with the bot that made it. Can others guess which bot drew which picture?

Coming up in the club

What will we build next?



Robot cars
that avoid walls and follow lines



Robot arms
that copy your hand



Smart devices
you control from your phone



Smart farms
that water plants by themselves



Safety systems
for fire, gas and security



AI projects
that see and recognise things



Next time we add a **brain** and **sensors**, and our machines become real robots!

Show what you know

What did we learn?

- [x] A robot senses, thinks and acts by itself
- [x] Robots have sensors, a controller, actuators, power, a body and a program
- [x] A DC motor is an actuator: it turns electricity into spinning
- [x] An off-centre weight turns spinning into shaking (vibration)
- [x] Our Art Bot only acts, so it is a machine, not yet a robot

Art Bot Builder
badge unlocked!

At home
Find 3 things at home that vibrate. Which ones have a spinning motor inside?

Exit ticket: what would you add to turn your Art Bot into a real robot?